**Mohammed Farhaan Ahmed**
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Portfolio: https://frhn.my.canva.site/
GitHub: https://github.com/elezenryfe

## PERSONAL PROFILE

A post-graduate and aspiring data scientist with an interest in data pre-processing, cleaning, analysing, and modelling. Working as a Data Analyst at BMW.

Relevant experience in Python Programming, SQL, R, and use of Microsoft tools (Excel, Word, Power BI). 2.5 years of experience as a data analyst/scientist at BMW with multiple placements across the western world. Detail-oriented and a critical thinker with excellent verbal and written communication skills. Numerically minded with a diverse experience in statistics, mathematical modelling, machine learning, and problem-solving. Have and willing to work both independently and as a supportive team member. Quick to adapt with an eagerness to learn new technologies.

## EDUCATION

**University of Leicester (First class)** Jan 2021 – May 2022

MSc. Data Analysis for Business Intelligence
Relevant Modules: Data Mining & Neural Networks, Fundamentals of Data Science, Introduction to R, Financial Services Information Systems and Linear Statistical Models.

**Jawaharlal Nehru Technological University, Hyderabad India (First class)** Aug 2015 – Jun 2019

Bachelor of Engineering (Computer Science) – Achieved distinction
Modules: Engineering Mathematics, C Programming, Algorithms, Data Structures, DBMS, OS, Web Technologies, Cloud Computing.

## RELEVANT WORK EXPERIENCE

**Data Analyst, BMW Financial Services** April 2024 – current

- Developed an automated testing tool saving 70% of the time during release of new products into AWS warehouse
- Developed end-to-end traditional AI model for predicting customer renewability in Financial Services
- Created a masking tool using AWS and open source services to obfuscate all personal and sensitive customer data before being processed by AI models

**Global Leadership Development Programme** Oct 2022 – April 2024

- **The graduate scheme is divided into four placements**

*Placement 1: Data Scientist, BMW Group UK (Farnborough)*

- Data extraction and collection from Athena database in AWS
- Performed data cleaning and pre-processing using SQL and Python.
- Built Machine Learning model for various use cases and used inferential statistics to evaluate models.
- Use of Sagemaker, EC2 instances, S3 buckets, and other amazon web services.

*Placement 2: Data Analytics, BMW Group AG (Munich).*

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- Worked on a new model for selling cars
- Developed an elasticity model to accurately price cars with highest optimization

*Placement 3: Data Analyst, BMW Group UK (Farnborough)*

- Automated performance reports that are required to be sent to headquarters monthly.
- Collected and ingested data into the warehouse to build a customer dashboard on Tableau

*Placement 4: Data Scientist, BMW Group US (New Jersey)*

- Created an AI detection model that successfully compared paper contracts against system values
- Increased efficiency as each manual comparison took 15 mins and presented AI model compared hundreds of contracts within seconds.

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**Data Curator/Scientist and Python Developer**, **INI Labs Inc.** Sept 2019 - Dec 2020

- Building HTML webpages of complex PDF, DOC, and TXT files using REGEX in Python.
- Performed data cleaning and pre-processing on unstructured files scraped from the internet.
- Linked webpages and rendered as a RESTful API using FLASK web framework, for prototype debugging in the development phase before migrating to the production environment.
- Oversaw and collaborated with team members for ensuring code optimization and code reviews.
- Biweekly presentation of code improvements and redundancies within the program.
- Worked on a model to automate the process and built a front-end application for accessing the converted HTML files.

**IT SKILLS:**

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Python, HTML/CSS/SCSS, JavaScript, Django, Flask, Java, SQL, C Language, Git, Tableau, Numpy, Pandas, Scikit, Oracle, APEX, Postgre SQL, AWS Athena, EC2, S3, Sagemaker, Comprehend

**Academic Projects:**

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- **Building predictive models for inpatient length-of-stay:** The ideal model was found to be the calibrated random forest, yielding accuracy and f1 scores of 79.80% and 79.71% respectively for multiclass variable prediction and 92.97% and 92.95% respectively for binary-class variable prediction. For additional predictive models (waiting time of patient, likelihood of rehospitalization) and related works, ***click here***
- **Building an EA simulator:** Created a minimum viable product that emulates the EA simulation feature in FIFA(21), and predicts the outcome of a match more realistically. Upon simulating an entire premier league season multiple times, the created simulator yielded an average accuracy of 85.5% in comparison to the real-world standings of the team. For additional models and related works, ***click here***
- **COVID analysis:** Performed EDA on the COVID cases of Kuwait, UAE, and Oman from its origin until mid-2021. Used analysis to determine the exponential and logistic growths of all waves of each country. Used SIR models to model a prediction on the different waves of each country. Modified model by adding parameters related to behavioural psychology. For detailed description of the models, ***click here***
- **Stock time series analysis:** Data evaluation and pre-processing of four stocks with their time-series data were collected from the Yahoo! Finance website: Baron Oil Plc (BOIL.L), Premier African Minerals Limited (PREM.L), Iconic Labs Plc (ICON.L), Lloyds Banking Group Plc (LLOY.L). Segmented the data at various tolerance levels using bottom-up piecewise linear segmentation and employed the "next day forecast" prediction. For detailed description, ***click here***
- For more projects, visit GitHub

**Languages:** English (Fluent), Urdu (Native), Hindi (Native), Arabic (Basic)

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